Deployment Diagram

«external»
User-Client
[laptop/desktop]
restic + SSH key + Tailscale

HTTPS :443 (public internet)
Let's Encrypt cert

joins after bootstrap
(pre-auth key, CL-F-04)

Public VPS
(dedicated, NET-F-01)

«public»
Entry-Hub
[Go]
HTTPS :443

Public VPS
(dedicated, NM-F-14)

«public»
Headscale
coordination + admin API

coordinates only
(cannot itself be a member, NM-F-12)

Proxmox VE Host

LXC Containers

«docker»
«service»
Security-Switch
[Go]

«docker»
«certAuthority»
Certificate-Authority
[smallstep/certificates]

«docker»
«infra»
MQTT-Broker
[Mosquitto]

«docker»
«infra»
Metrics-Collector
[Go]

«docker»
«infra»
Metrics-Visualizer
[Grafana]

KVM Virtual Machines
(RNF-ORG-04)

«docker»
«service»
Database-Vault
[Go]

«docker»
«service»
Storage-Service
[Go, SFTP]

«docker»
«service»
Network-Manager
[Go]

REST API :443 (mTLS + token)
pre-auth key / ACL policy

Headscale Mesh VPN
private network, mTLS
certs from private CA, TLS 1.3

Entry-Hub and Headscale are the only publicly reachable nodes (NET-F-01).
Every other node communicates exclusively over
the mesh, mutually authenticated with private-CA certificates.